Please submit your solutions to the following problems on Gradescope by **6pm** on the due date. Collaboration is encouraged, however, you must write up your solutions individually.

1) Polynomial Solutions. Using a power series ansatz, show that for each $n = 1, 2, 3, \ldots$, the following differential equations have a polynomial solution of degree n.

(a) The Hermite differential equation: $y'' - 2xy' + 2ny = 0$.

(b) The Laguerre differential equation: $xy'' + (1 - x)y' + ny = 0$.

2) Modified Power Series. For each of the following ordinary differential equations, identify singular points (if any) and derive a series solution that is valid for $-1 \leq x \leq 1$. Present your final solution in a simple form when possible.

(a) $(x^2 + 1)y'' - xy' + y = 0$

(b) $x^2y'' + (x + 1)y' - y = 0$

(c) $x^2y'' + x^2y' - 2y = 0$

(d) $(x - 1)y'' - xy' + y = 0$